

Executive Goal: Deliver a production-grade, mathematically defensible, zero-placeholder end-to-end pipeline transforming single optical RGB remote-sensing imagery into calibrated relative/absolute Digital Surface Models (DSM), rendered in a high-performance interactive 3D WebGL flythrough environment with spatial structural measurements.

1. Dual Ingestion Pipelines

Mode 1: Non-Georeferenced (PNG/JPG)

- **Input:** Standard optical aerial / satellite RGB photograph.
- **Inference:** Zero-shot Depth Anything V2 monocular backbone.
- **Output:** Relative Digital Surface Model (rDSM) normalized [0, 1].
- **3D View:** Height-exaggerated terrain mesh textured with source RGB.
- **Integrity:** UI explicitly states heights are relative (non-metric).

Mode 2: Georeferenced (GeoTIFF Raster)

- **Input:** GeoTIFF raster preserving CRS, bounding box & GSD.
- **Alignment:** Matched SRTM/COP30 reference DEM or user GCPs.
- **Calibration:** Robust Huber regression ($Z = s \cdot D_{rel} + t$).
- **Output:** Absolute Metric DSM (meters) + GeoTIFF export.
- **Validation:** Displays MAE, RMSE, Pearson r, and R² accuracy.

2. Depth-to-Elevation Mathematical Calibration

For georeferenced imagery, physical metric heights are determined via robust M-estimation against reference DEM samples / Ground Control Points, rejecting outlier vegetation/cloud artifacts:

Affine Metric Mapping: $Z_{pred}(x, y) = scale \cdot D_{rel}(x, y) + offset$
Robust Huber Loss: $\min_{\{s, t\}} \sum p_{\delta}(Z_{ref}(x_i, y_i) - (s \cdot D_{rel}(x_i, y_i) + t))$
Validation Metrics: $RMSE = \sqrt{\frac{1}{N} \sum (Z_{pred} - Z_{ref})^2}$ | $MAE = \frac{1}{N} \sum |Z_{pred} - Z_{ref}|$ | $R^2 = 1 - (SS_{res} / SS_{tot})$

3. Full-Stack System Architecture

Layer	Technologies	Key Functional Responsibilities
Backend API	FastAPI, Uvicorn, Python 3.10+	RESTful endpoints, pipeline orchestration, GeoTIFF serialization, analytics.
ML Core	PyTorch / ONNX, Depth-Anything-V2	Zero-shot monocular depth prediction with GPU acceleration & CPU fallback.
Geospatial	rasterio, GDAL, affine, pyproj, scipy	CRS reprojection, DEM resampling, GCP indexing, Huber regression.
Frontend UI	React 19, TypeScript, Tailwind CSS	Synchronized 4-pane comparative viewer, inspection panel, and controls.
3D Engine	Three.js, WebGL, BufferGeometry	Drone first-person fly navigation, height exaggeration, optical texture mapping.

4. Interactive 3D Terrain Engine & Geospatial Analysis Suite

3D Terrain Engine Capabilities

- **Dual Camera Controls:** Aerial Drone First-Person (WASD + Mouse) & Smooth Orbit mode.
- **Real-time Exaggeration:** Vertical scale slider (0.1x to 5.0x) for subtle topographical features.
- **Dynamic Shading:** True optical RGB texture, wireframe overlay, and scientific colormaps.
- **Sun Angle Simulation:** Configurable solar azimuth and elevation to reveal structural relief.

Geospatial Analysis Tools

- **Point Query:** Interactive raycasting for instantaneous (Lat, Lon, Elevation Z) readouts.
- **2-Point Measurement:** Structural height differences (ΔZ) and 3D Euclidean distances.
- **Slope Computation:** Surface grade percentage and pitch angle (θ°) between points.
- **Cross-Section Profile:** Dynamic ridge-line elevation profile chart across transects.
- **Hypsometric Stats:** Elevation histogram, Min/Max/Mean, surface area estimation.

5. Modular Project Directory Structure

```
DepthWizard/  
■■■ README.md, ARCHITECTURE.md, MODEL_DECISIONS.md, CALIBRATION.md, DEMO_GUIDE.md  
■■■ backend/  
■ ■■■ app/ [main.py, config.py]  
■ ■ ■■■ api/ [routes_upload.py, routes_process.py, routes_mesh.py, routes_analysis.py, routes_demo.py]  
■ ■ ■■■ ml/ [base_model.py, depth_anything_v2.py, model_manager.py, colormaps.py]  
■ ■ ■■■ geospatial/ [geotiff_handler.py, dem_aligner.py, calibrator.py, gcp_manager.py]  
■ ■ ■■■ processing/ [pipeline.py, mesh_generator.py, exporter.py]  
■ ■■■ tests/ [test_geotiff_crs.py, test_calibration.py, test_mesh_generation.py, test_end_to_end.py]  
■■■ frontend/  
■ ■■■ src/ [App.tsx, components/MultiPaneViewer.tsx, components/TerrainCanvas.tsx, three/SceneManager.ts]  
■■■ data/demo/ [Sample GeoTIFFs, SRTM 30m reference DEMs, GCP ground-truth records, Optical RGBs]
```

6. 3-Minute SIH Jury Presentation Script

1. **Ingestion:** Upload satellite GeoTIFF. System extracts spatial bounds (EPSG:32643, GSD: 0.5m).
2. **Monocular Depth:** Depth Anything V2 generates dense relative depth preserving structural edges.
3. **DEM Calibration:** Huber regression aligns with SRTM DEM, reporting real accuracy ($R^2 = 0.94$, RMSE: 2.8m).
4. **3D Texture Mapping:** Generates triangulated terrain mesh textured 1:1 with optical RGB imagery.
5. **Flythrough & Analysis:** Pilot drone flight, measure structural height (ΔZ), and export GeoTIFF DSM.

7. Acceptance Checklist & Verification Criteria

Component	Acceptance Benchmark	Verification Strategy
Dual Format Support	PNG/JPG (relative) & GeoTIFF (absolute)	Automated rasterio metadata & CRS validation tests.
Depth Inference	< 2.5s on GPU / < 8s on CPU	Benchmarked on standard 1024x1024 optical tiles.
Calibration Quality	$R^2 > 0.85$, RMSE < 5m on test DEM reference	Residual error verification on held-out control points.
3D Rendering	60 FPS WebGL rendering with dynamic LOD	Three.js frame rate profiling & memory leak checks.
Offline Reliability	100% operational without external internet	Bundled demo datasets with pre-cached assets.